Mean, Median, and Baselines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

TEACHER GUIDE · 1 OF 9

Mean, Median, and Baselines

> Prepare Mean, Median, and Baselines: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Fit **mean** and **median constant baseline**s on **training** targets

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish numeric **prediction** from classification

Reference example, not the only acceptable wording: Teacher explanation: Residuals -1, 1, 0 cm; the first record is overpredicted.

Precision and Recall

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Look for: Choose the denominator that answers the question

Reference example, not the only acceptable wording: Teacher explanation: Predicted positives TP+FP=7; reference positives TP+FN=4.

Scores and Calibration Intuition

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Interpret calibration using groups of forecasts

Reference example, not the only acceptable wording: Teacher explanation: 2/10=.2 and 6/10=.6.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

Concept to explain

Mean and **median** are transparent regression baselines; **outlier**s affect them differently.

Term	Meaning
mean	sum divided by count
median	middle ordered value
outlier	value far from most observations
location baseline	single typical value predicted for all cases

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

> Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on constant **prediction** baselines. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Opposite signs can cancel; a magnitude-based loss avoids that cancellation.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain **mean**, **median**, and baselines using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Mean, Median, and Baselines using the supplied case, labelled mechanism and

> reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

Question to explain

QUESTION Choose and justify a baseline under an extreme value.

MODEL IDEA **Mean** and **median** are transparent regression baselines; **outlier**s affect them differently.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Heights are 10, 11, 12, 13, and 54 cm. **Mean** = 100 ÷ 5 = 20 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

Step 1

STEP 1 Heights are 10, 11, 12, 13, and 54 cm.

Step 2

STEP 2 **Mean** = 100 ÷ 5 = 20 cm.

Step 3

STEP 3 **Median** is 12 cm, the middle ordered value.

Step 4

STEP 4 For typical-height **absolute error**, **median** is more robust; still report why 54 is present.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

The teaching case

Training heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

The teaching case

Compute both and **training absolute error**s.

Exact explanation to model

Why the result follows

Mean=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

Explain the added worked example

Explain the added worked example · 1

Before a complex regression model, fit a simple constant predictor. A **mean** baseline uses the **training** average. A **median** baseline uses the middle **training** target.

They ignore features but provide a comparison that is easy to reproduce. Evaluation targets must not determine the constants.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines constant **prediction** baselines.

Training targets [2, 4, 6, 12] have **mean** 24/4=6 and **median** (4+6)/2=5. On separate targets [4, 8], the **mean** baseline predicts [6, 6] with **absolute error**s [2, 2], averaging 2.

The **median** baseline predicts [5, 5] with errors [1, 3], also averaging 2.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Key terms

Term	Meaning
Constant baseline	One fixed prediction for every query
Training mean	Sum of training targets divided by their count
Training median	Middle value of sorted training targets
Absolute error	Magnitude of observed minus predicted

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Constant prediction baselines at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training targets	Fit mean 6	Freeze constant
Training targets	Fit median 5	Freeze constant
Evaluation targets	Compare fixed predictions	Both MAE 2 here

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

Calculate MAE for both baselines on [4, 8]. Explain the steps or evidence behind your response.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Mean, Median, and Baselines

Fit two constant height baselines and compare them on held-out examples.

The facts we will use

> **Supplied facts**
>
> **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

> **Supplied facts**
>
> All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/20

Mean, Median, and Baselines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Fit mean

Training mean=(2+4+6+20)/4=8 cm.

2. Fit median

Training median=(4+6)/2=5 cm.

3. Evaluate constants

Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm.

CHANGE ONE THING

Replace training 20 with 8 while keeping evaluation unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit mean	Training mean=(2+4+6+20)/4=8 cm.
2. Fit median	Training median=(4+6)/2=5 cm.
3. Evaluate constants	Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Mean and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

Change one thing in the pictured case

Change one thing

Change: Replace **training** 20 with 8 while keeping evaluation unchanged.

Predicted result

Prediction: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

Why the result follows

Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Mean, Median, and Baselines, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: **Training** heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

2. State the decision to explain

Write the question to explain: Compute both and **training absolute error**s.

3. Reveal the worked reasoning

Reveal after **prediction**s: **Mean**=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

Hold back the independent task

Keep for an independent attempt: For **training** heights 2, 3 and 10 cm, calculate **mean** and **median**. Which constant is less influenced by the high value?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the

result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: Both score 2; their per-record **absolute error**s differ. Exit response: It fits using evaluation answers and makes the baseline unavailable for genuinely unseen cases.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Why can signed residuals hide error size? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify constant baseline in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: fit two constants; score separately. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

> **Stage:** Session 2 start to 8 minutes
>
> **Teacher action:** Retrieve the distinction between constant baseline and training mean. Invite a counterexample to an overconfident claim.

> **Stage:** 8 to 20 minutes
>
> **Teacher action:** Discuss the application and complete W 3: Replace 12 by 40. What changes?

> **Stage:** 20 to 35 minutes
>
> **Teacher action:** Collect W4 independently before giving hints. Assess these outcomes: Fit mean and median constant baselines on training targets; Compare constant predictions on separate records. Record any support used.

> **Stage:** 35 to 45 minutes
>
> **Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion practice

Completion task: Calculate MAE for both baselines on [4, 8].

Reasoning to listen for

Teacher check: Teacher explanation: Both score 2; their per-record **absolute error**s differ.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Constant **prediction** baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> ### Expected repair
>
> Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: **Mean** 6; **median** 5.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No, both have3 cm MAE on the supplied evaluation set; do not claim a strict winner from the **training** statistic alone. Changed-case check: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

Mean is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Changed-case reasoning: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Mean, Median, and Baselines investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. Baseline Tournament: compute **mean** and **median**, predict every case, compare MAE, and explain the **outlier**'s effect.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR The **mean** is always the fairest and most scientific summary.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: **Mean** 6; **median** 5.

A stronger claim makes a stronger project: It fits using evaluation answers and makes the baseline unavailable for genuinely unseen cases.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Mean, Median, and Baselines questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Mean=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

Independent question

Independent question · withhold the key

For **training** heights 2, 3 and 10 cm, calculate **mean** and **median**. Which constant is less influenced by the high value?

Expected independent answer

Teacher key · Expected independent answer

Mean is 15/3 = 5 cm; **median** is 3 cm. The **median** is less affected by the high value. Fit a baseline from **training** data only.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “**Mean** and **median** must always be equal.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Mean=5, **median**=3. **Mean** absolute-error total=4+2+6=12; **median** total=2+0+8=10. These are **training** results. Fit the constant using **training** targets and measure final performance on untouched test targets.

Explanations for the additional workbook examples

W1 Fit two constants

Workbook question · W1

Find the **mean** and **median** of the worked **training** targets.

Teacher reasoning · W1

Teacher explanation: **Mean** 6; **median** 5.

Teacher reasoning · W1

For the **mean** baseline, sum the worked **training** targets and divide by their count to obtain 6. For the **median** baseline, sort the **training** targets and take the middle value, 5.

Freeze these choices before examining evaluation targets; choosing a new constant from evaluation outcomes would contaminate the comparison.

W2 Score separately

Workbook question · W2

Calculate MAE for both baselines on [4, 8].

Teacher reasoning · W2

Teacher explanation: Both score 2; their per-record **absolute error**s differ.

Teacher reasoning · W2

For constant 6, **absolute error**s on targets 4 and 8 are 2 and 2, so MAE is (2+2)/2 = 2. For constant 5, the errors are 1 and 3, so MAE is (1+3)/2 = 2.

Equal averages can hide different per-record errors; retain the error table as well as the summary.

W3 Inspect an extreme value

Workbook question · W3

Replace 12 by 40. What changes?

Teacher reasoning · W3

Teacher explanation: **Mean** becomes 13; **median** stays 5. Investigate validity before deciding treatment.

W4 Apply independently

Workbook question · W4

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

Teacher reasoning · W4

Teacher explanation: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3.

W5 Prevent leakage

Workbook question · W5

Why not predict the evaluation **mean** as the baseline?

Teacher reasoning · W5

Teacher explanation: It fits using evaluation answers and makes the baseline unavailable for genuinely unseen cases.

Assess independent transfer

Independent question · withhold the key

Selected case: **Training** [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

> **Teacher key · Selected case answer**
>
> Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.
>
> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Mean, Median, and Baselines to Responsible Plant Classifier, adapt support and plan a later changed-case check.

Responsible Plant Classifier

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 21: **Absolute Error** and MAE. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **constant baseline** and **training mean** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: **Mean** 6 and **median** 4. **Mean** errors [3, 3] MAE 3; **median** errors [1, 5] MAE 3. Fresh retry **training** [2, 2, 8], evaluation [1, 7]: **mean** 4 **median** 2; MAE 3 for each. Require **training**-only **fitting**.

Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Absolute and squared loss. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/21, ai-4/23, ai-4/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Mean, Median, and Baselines lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Fit **mean** and **median constant baseline**s on **training** targets

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

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> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Calculate MAE for both baselines on [4, 8].

Supported practice answer

Reasoned practice answer: Teacher explanation: Both score 2; their per-record **absolute error**s differ.

Check every learner before discussion: Use the worked case for Constant **prediction** baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: **Mean** 6; **median** 5.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Replace **training** 20 with 8 while keeping evaluation unchanged. Require a **prediction** and a reason before comparing. Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

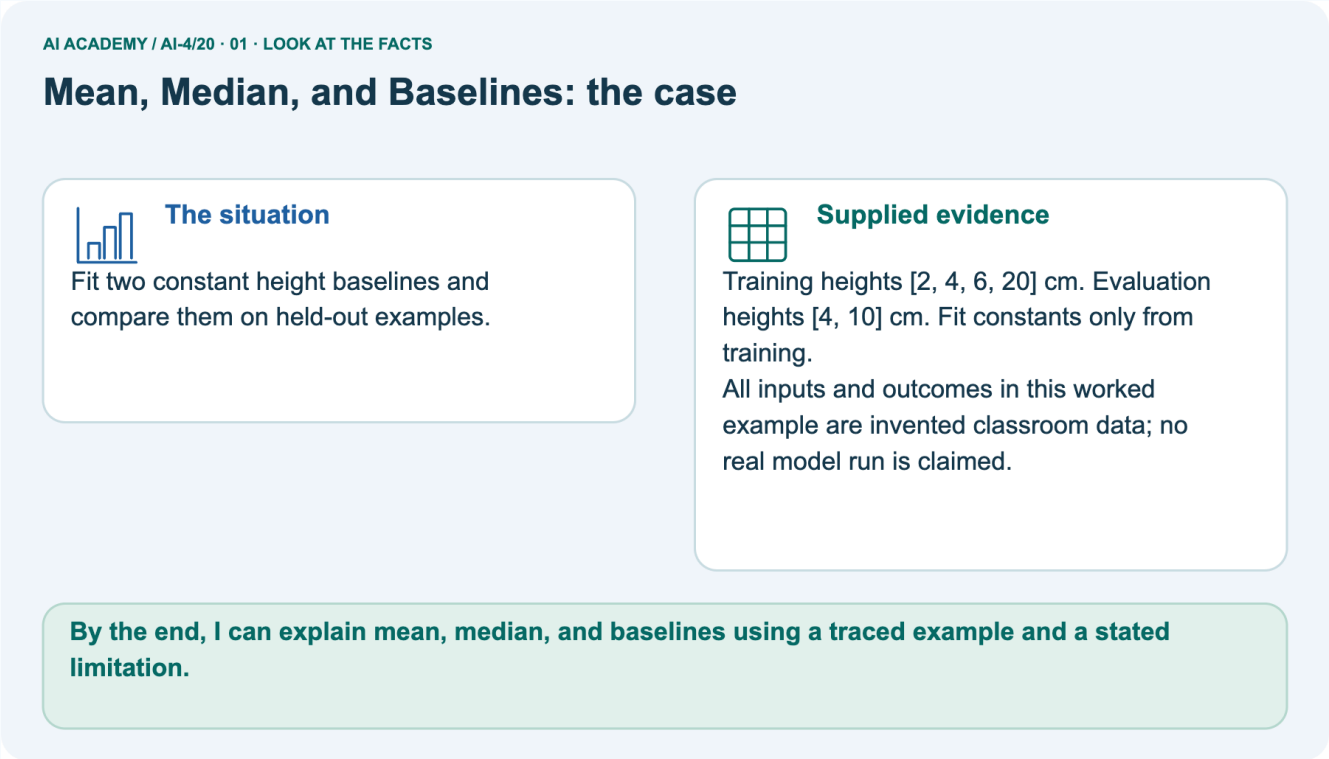
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Mean, Median, and Baselines graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Mean, Median, and Baselines: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit two constant height baselines and compare them on held-out examples. Supplied facts: **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

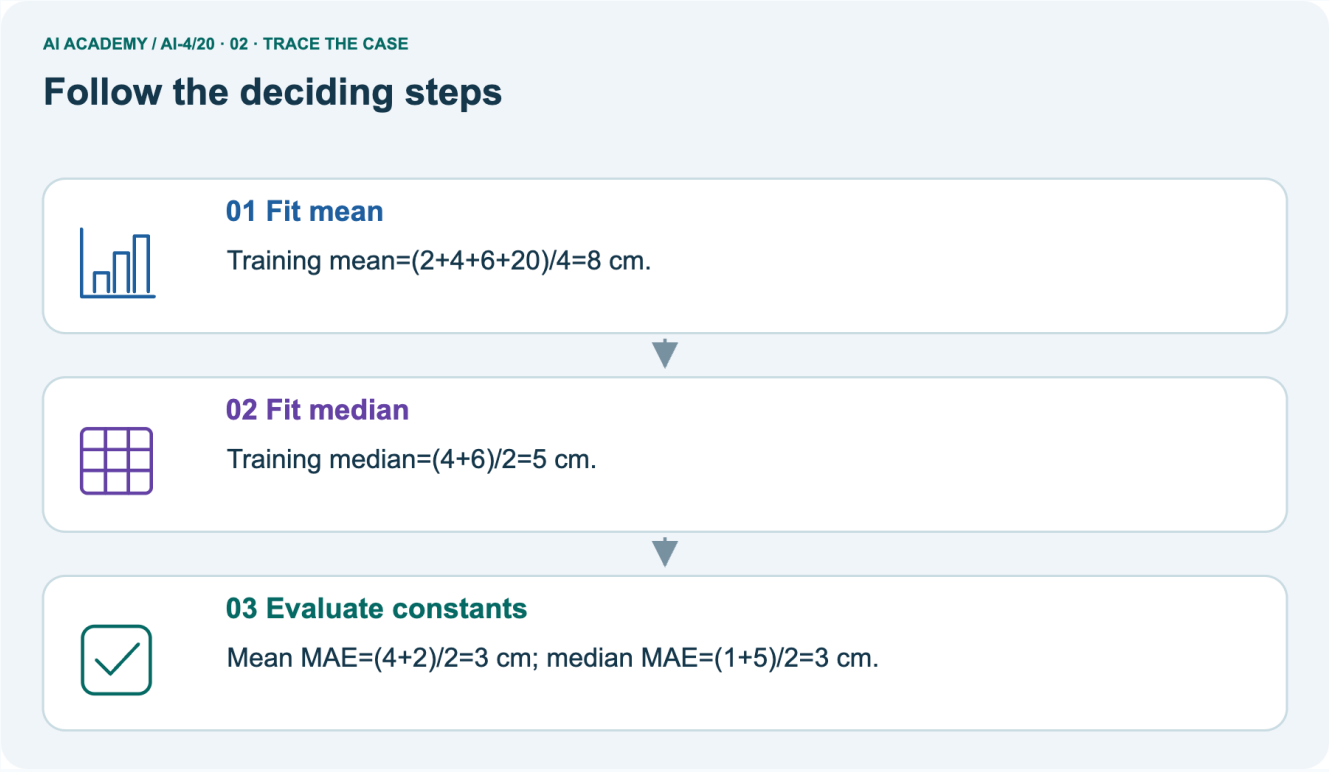
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **mean**: **Training mean**=(2+4+6+20)/4=8 cm. → Fit **median**: **Training median**=(4+6)/2=5 cm. → Evaluate constants: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Constant prediction baselines at a glance

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Constant prediction baselines at a glance

Part	Evidence or action	Interpretation
Training targets	Fit mean 6	Freeze constant
Training targets	Fit median 5	Freeze constant
Evaluation targets	Compare fixed predictions	Both MAE 2 here

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

Follow the labelled picture with the matching explanation. Open the vector diagram

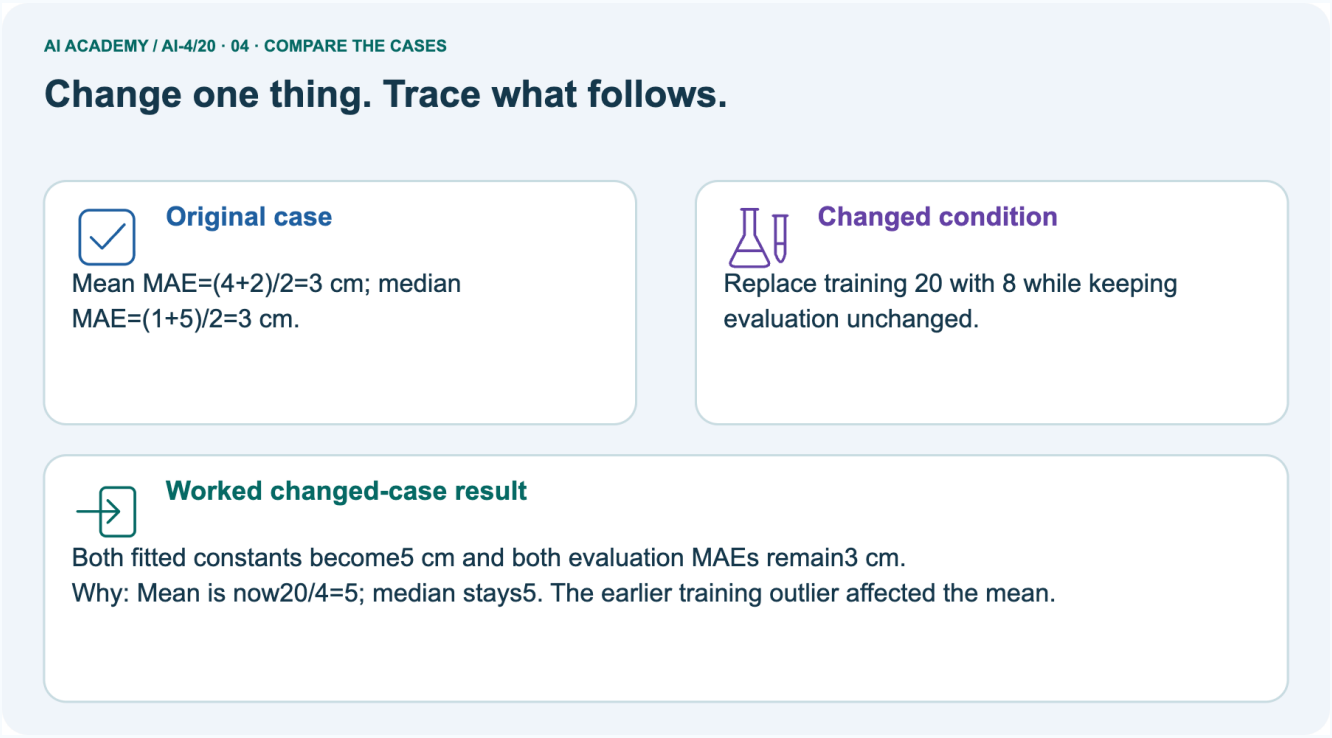
Constant **prediction** baselines at a glance. **Training** targets | Fit **mean** 6 | Freeze constant. **Training** targets | Fit **median** 5 | Freeze constant. Evaluation targets | Compare fixed **prediction**s | Both MAE 2 here

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Changed condition: Replace **training** 20 with 8 while keeping evaluation unchanged.

Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

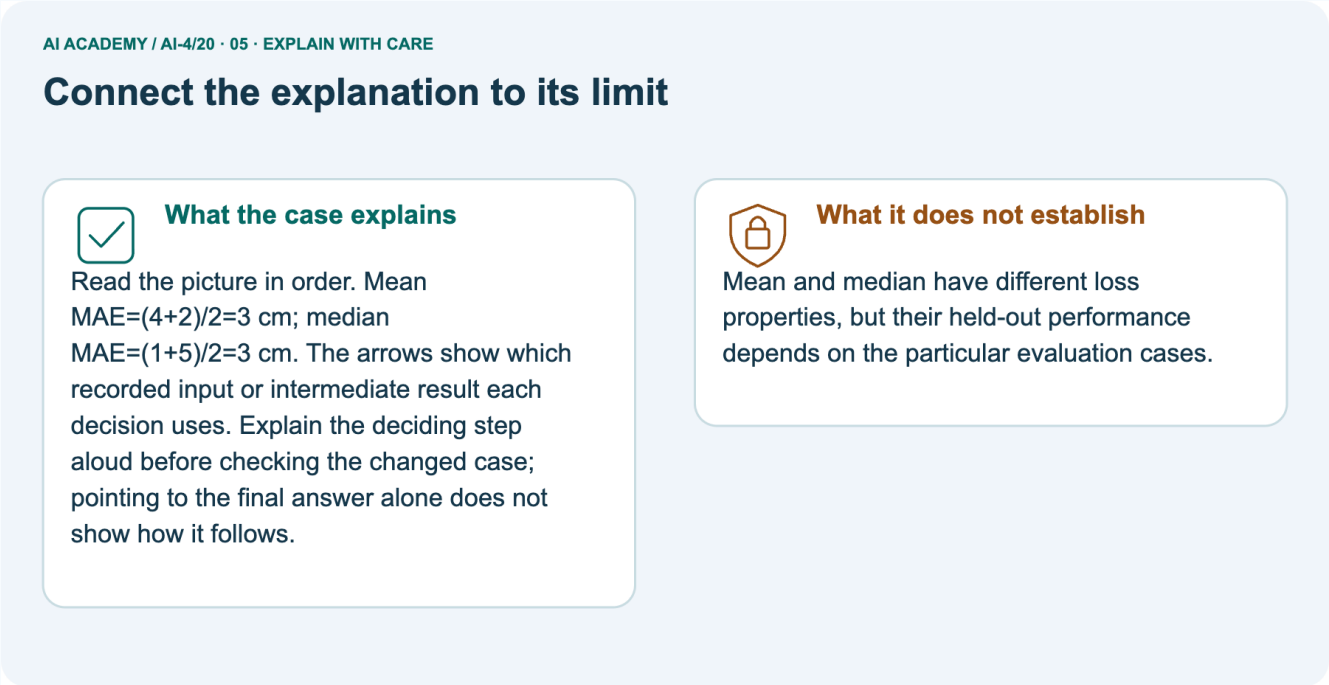
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

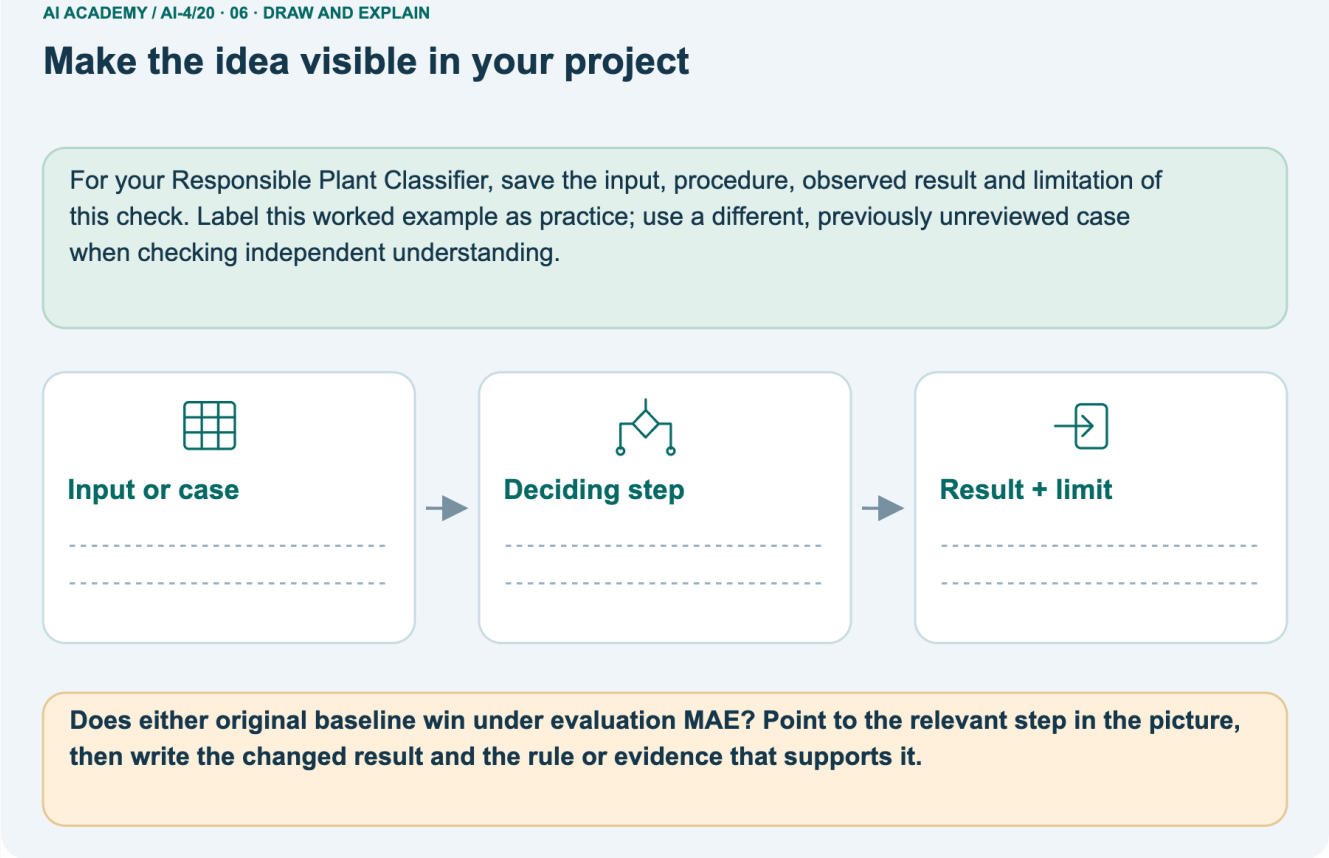
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No, both have3 cm MAE on the supplied evaluation set; do not claim a strict winner from the **training** statistic alone.

Changed-case check: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.